

**Risk Parity** is the strategy du jour and chooses asset weights proportional to the inverse of variance [*Risk Parity (Variance)*] or to the inverse of volatility [*Risk Parity (Volatility)*]. The term “risk parity” was originally coined by Edward Qian in 2005.<sup>27</sup> It has shot to prominence in the practitioner community because of the huge success of Bridgewater Associates, a large hedge fund with a corporate culture that has been likened to a cult.<sup>28</sup> Bridgewater launched the first investment product based on risk parity called the “All Weather” fund in 1996. In 2011 the founder of Bridgewater, Ray Dalio, earned \$3.9 billion.<sup>29</sup> (I cover hedge funds in chapter 17). Bridgewater’s success has inspired many copycats. The original implementations of risk parity were done on variances, but there are fans of weighting on volatilities.<sup>30</sup>

**Minimum Variance** is the portfolio on the left-most tip of the mean-variance frontier, which we’ve seen before.

**Equal Risk Contributions** form weights in each asset position such that they contribute equally to the total portfolio variance.<sup>31</sup>

**Kelly (1956) Rule** is a portfolio strategy that maximizes the expected log return. In the very long run, it will maximize wealth. (I explain more in chapter 4.)

**Proportional to Sharpe Ratio** is a strategy that holds larger positions in assets that have larger realized Sharpe ratios over the last five years.

Over this sample a 100% investment in U.S. equities had a Sharpe ratio of 0.35. This is dominated by all the diversified portfolios in the horserace, except for the most unconstrained mean-variance portfolio. This is consistent with the advice from the example in section 2 where no one should hold a 100% U.S. equity portfolio; diversification produces superior portfolios with lower risk and higher returns.

Table 3.14 reports the results of the horserace. Mean-variance weights perform horribly. The strategy produces a Sharpe ratio of just 0.07, and it is trounced by all the other strategies. Holding market weights does much better, with a Sharpe ratio of 0.41. This completely passive strategy outperforms the Equal Risk Contributions and the Proportional to Sharpe Ratio portfolios (with Sharpe ratios of 0.32 and 0.45, respectively). Diversity Weights tilt the portfolio toward the asset classes with smaller market caps, and this produces better results than market weights.

<sup>27</sup>Qian, E., 2005, Risk Parity Portfolios: Efficient Portfolios through True Diversification, PanAgora.

<sup>28</sup>Kevin Roose, “Pursuing Self-Interest in Harmony with the Laws of the Universe and Contributing to Evolution is Universally Rewarded,” *New York Magazine*, Apr. 10, 2011.

<sup>29</sup>I cover hedge funds in chapter 17.

<sup>30</sup>Versions of risk parity where the weights are inversely proportional to volatility are advocated by Martellini (2008) and Choueifaty and Coignard (2008).

<sup>31</sup>See Qian (2006) and Maillard, Roncalli, and Teiletche (2010).